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GitHub Link	https://github.com/192425292simats-byte/MLA0403-DEEP-LEARNING

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 3

Technical Presentation

COURSE NAME: Deep Learning
SLOT :A

COURSE CODE: MLA0403
Faculty : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35%
Assessment Tool Weightage: 50%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I (V Sunil Kumar / 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student:V Sunil

Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (3)	Accuracy of Solution (2)	Clarity (1)	Total (10)
Weightage	20 %	30%	30%	20 %	10 %	100%
Q1						
Total						

Signature of the Faculty:

Total Marks:

PHOTO :



QUESTIONS:

PRESENTATION ANSWER :

Deep Belief Networks and Boltzmann Machines

1. Introduction

Deep Belief Networks (DBNs) and Boltzmann Machines (BM) are generative deep learning models that learn patterns and representations from data. They use **visible units** to represent input data and **hidden units** to learn important features. Both are based on **energy-based learning**, where the model assigns low energy to likely data patterns and high energy to unlikely patterns.

2. Boltzmann Machine

A **Boltzmann Machine (BM)** is a stochastic neural network consisting of visible and hidden units.

Architecture:

- **Visible units:** Receive and represent input data.
- **Hidden units:** Learn relationships and features from the input.
- Units are connected using weighted connections.
- The connections are generally undirected.

Working:

1. Input data is given to the visible units.
 2. Visible and hidden units interact with each other.
 3. The network calculates an energy value for different configurations.
 4. The model adjusts its weights to give lower energy to correct data patterns.
 5. After training, it can generate or reconstruct similar data.
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3. Energy-Based Learning

Boltzmann Machines use an **energy function** to represent the relationship between visible and hidden units.

A simple energy function is:

$$E(\mathbf{v}, \mathbf{h}) = -\sum w_{ij} v_i h_j - \sum a_i v_i - \sum b_j h_j$$

Where:

- **v** = visible units
- **h** = hidden units
- **w** = connection weights
- **a, b** = biases

The objective is to assign:

- **Low energy** → likely/correct patterns
- **High energy** → unlikely patterns

The weights are updated during training to improve the probability of observed data.

4. Deep Belief Network

A **Deep Belief Network (DBN)** is a generative neural network formed by stacking multiple layers of learning units, commonly **Restricted Boltzmann Machines (RBMs)**.

Architecture:

Input → Hidden Layer 1 → Hidden Layer 2 → Hidden Layer 3 → Output/Representation

- **Visible layer:** Represents input data.
- **Hidden layers:** Learn increasingly complex features.
- Lower layers learn simple features.
- Higher layers learn more abstract features.

For example, in image recognition:

- First layer → edges
- Second layer → shapes
- Third layer → objects or complex patterns

5. Training Process of DBN

DBNs are commonly trained using **layer-wise unsupervised pre-training**.

1. The input data is given to the first RBM.
2. The first RBM learns basic features.
3. The output of the first layer becomes the input to the next layer.
4. Each additional RBM learns higher-level features.
5. After pre-training, the entire network can be fine-tuned for a specific task.

This approach helps the network learn useful representations even when labeled data is limited.

6. DBN vs Boltzmann Machine

Feature	Boltzmann Machine	Deep Belief Network
Structure	Single/stochastic network	Multiple stacked layers
Hidden Units	One or more hidden units	Multiple hidden layers
Connections	Generally fully connected and undirected RBM-based layers	
Learning	Energy-based learning	Layer-wise RBM training
Feature Learning	Learns basic relationships	Learns hierarchical features
Training Complexity	High	More manageable
Main Use	Generative modeling	Feature extraction and classification
Deep Learning	Limited depth	Designed for deep architectures

7. Feature-Learning Capability

Boltzmann Machine:

- Learns relationships between input variables.
- Hidden units capture dependencies in the data.
- Useful for learning probability distributions.

Deep Belief Network:

- Learns features hierarchically.
 - Converts raw data into increasingly meaningful representations.
 - Useful for complex data such as images, speech, and text.
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8. Real-World Application: Image Recognition

DBNs and Boltzmann Machines can be applied to **handwritten digit recognition**.

For example, consider the MNIST dataset:

Input Image → Visible Units → Hidden Layer 1 → Hidden Layer 2 → Digit Representation

- Visible units receive pixel values.
- The first hidden layer learns edges and basic patterns.
- Higher hidden layers learn shapes and digit structures.
- The learned representation can be used to classify digits from **0 to 9**.

DBNs are particularly useful because they can automatically learn important features instead of requiring all features to be manually designed.

9. Advantages

Boltzmann Machines

- Can model complex probability distributions.
- Learn features automatically.
- Useful for generative modeling.

Deep Belief Networks

- Learn hierarchical representations.
 - Useful for high-dimensional data.
 - Can work effectively with unsupervised pre-training.
 - Useful for feature extraction and classification.
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10. Conclusion

Boltzmann Machines and Deep Belief Networks are important **generative learning models** based on energy-based learning. A Boltzmann Machine focuses on learning relationships between visible and hidden units, while a DBN extends this concept by stacking multiple layers to learn hierarchical features. DBNs are especially useful for applications such as **image recognition, speech processing, and pattern recognition**, where automatic feature learning is important.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding ; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

